

SAT Unit 3.1

Ratios, Rates, and Units

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Overview

- 1 Ratio
- 2 Rates and Units
- 3 Proportions

Ratio

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What is a Ratio?

A **ratio** is a comparison between two numbers. It can be written in three different ways:

$$\frac{a}{b}, \quad a : b, \quad \text{or } a \text{ to } b$$

Example: The ratio of boys to girls in a class is 3 to 4. - This means for every 3 boys, there are 4 girls. - The fraction form is $\frac{3}{4}$.

Example: Growth Factor Calculation

Species of tree	Growth factor
River birch	3.5
Cottonwood	2.0
Black walnut	4.5
White birch	5.0
American elm	4.0
Pin oak	3.0
Shagbark hickory	7.5

One method of calculating the approximate age, in years, of a tree of a particular species is to multiply the diameter of the tree, in inches, by a constant called the growth factor for that species.

The table on the previous slide provides the growth factors for eight species of trees.

If a white birch tree and a pin oak tree each now have a diameter of 1 foot (12 inches), which of the following will be closest to the difference in inches of their diameters 10 years from now?

Example: Solution

Step 1: Calculate the diameter increase for each tree in 10 years

$$\text{Age} = \text{Diameter} \times \text{Growth Factor}$$

Correct Answer: 1.3

Rates and Units

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Rates and Unit Rates

A **rate** is a special type of ratio that compares different units.

Example: A car travels 180 miles in 3 hours. The rate is:

$$\frac{180 \text{ miles}}{3 \text{ hours}} = 60 \text{ miles per hour}$$

A **unit rate** simplifies the rate so that the denominator is 1.

Example: If 12 apples cost \$6, the unit price per apple is:

$$\frac{6 \text{ dollars}}{12 \text{ apples}} = 0.50 \text{ dollars per apple}$$

Unit Conversion

Key Idea: Multiply by a conversion factor equal to 1.

$$\text{Quantity in old unit} \times \frac{\text{new unit}}{\text{old unit}} = \text{Quantity in new unit}$$

Example 1: Inches to Feet

$$36 \text{ inches} \times \frac{\frac{1}{12} \text{ foot}}{1 \text{ inches}} = 3 \text{ feet}$$

Example 2: Minutes to Seconds

$$5 \text{ minutes} \times \frac{60 \text{ seconds}}{1 \text{ minute}} = 300 \text{ seconds}$$

Tips:

- Always write units clearly.
- Cancel the units you don't need.
- Use a conversion factor that equals 1.
- Check your final units!

Question: Unit Conversion (Square Miles to Square Yards)

A certain town has an area of 4.36 square miles. What is the area, in **square yards**, of this town?

Given: 1 mile = 1,760 yards

- A. 404
- B. 7,674
- C. 710,459
- D. 13,505,536

Answer Explanation: Unit Conversion

Step 1: Understanding the conversion factor We know that:

$$1 \text{ mile} = 1,760 \text{ yards}$$

Since we are dealing with square miles, we square both sides:

$$1 \text{ square mile} = (1,760 \text{ yards})^2 = 3,097,600 \text{ square yards}$$

Step 2: Multiply by given area The given area is 4.36 square miles.
Converting to square yards:

$$4.36 \times 3,097,600 = 13,505,536 \text{ square yards}$$

Correct Answer: D

Proportions

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Solving Proportions

A **proportion** is an equation stating that two ratios are equal:

$$\frac{a}{b} = \frac{c}{d}$$

Cross Multiplication: If $\frac{3}{4} = \frac{x}{12}$, solve for x :

$$3 \times 12 = 4 \times x$$

$$36 = 4x \quad \Rightarrow \quad x = 9$$

Question: Solving for n

Problem Statement:

If

$$\frac{4a}{b} = 6.7 \quad \text{and} \quad \frac{a}{bn} = 26.8,$$

what is the value of n ?

Answer Explanation: Solving for n

Step 1: Express a/b in terms of known values From the first equation:

$$\frac{4a}{b} = 6.7$$

Dividing both sides by 4:

$$\frac{a}{b} = \frac{6.7}{4} = 1.675$$

Step 2: Substitute into the second equation

$$\frac{a}{bn} = 26.8$$

Rewriting using $\frac{a}{b} = 1.675$:

$$\frac{1.675}{n} = 26.8$$

Final Answer:

0.0625

Practice Questions

Question 1: If the ratio of boys to girls in a school is 5:8, and there are 200 boys, how many girls are there?

Question 2: A car travels 240 miles in 4 hours. What is its speed in miles per hour?

Question 3: If $\frac{7}{9} = \frac{x}{36}$, solve for x .

Conclusion

- Ratios compare two quantities
- Rates involve different units, and unit rates have a denominator of 1
- Proportions are equations that show two equal ratios
- Use cross multiplication to solve proportions
- These concepts are frequently tested on the SAT!

Practice

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Ratio Word Problem

The ratio of blueberries to grapes in a fruit salad is 5 to 12. If there are x blueberries in the fruit salad, which expression represents the number of grapes in the fruit salad?

A. $\frac{5}{12}x$

B. $\frac{12}{5}x$

C. $\frac{12}{17}x$

D. $60x$

Answer Explanation

To maintain the ratio 5 : 12 (blueberries to grapes), we set up the proportion:

$$\frac{5}{12} = \frac{x}{g} \Rightarrow g = \frac{12}{5}x$$

Correct Answer: $\frac{12}{5}x$

Unit Conversion: Area

The area of Wompatuck State Park in Massachusetts is 5.5 square miles. Which of the following is closest to the area of this park, in **square kilometers**? (Use 1 kilometer = 0.62 mile.)

- A. 2.1
- B. 3.4
- C. 8.9
- D. 14.3

Answer Explanation

We are given:

$$\text{Area in square miles} = 5.5, \quad 1 \text{ kilometer} = 0.62 \text{ mile}$$

So,

$$1 \text{ square mile} = \left(\frac{1}{0.62} \right)^2 \approx 2.6 \text{ square kilometers}$$

$$5.5 \text{ square miles} \approx 5.5 \times 2.6 = 14.3 \text{ square kilometers}$$

Correct Answer: D 14.3

Scale Modeling

A model of a building has a scale where a width of 5 inches in the model represents an actual width of 16 feet. A room in the building has an actual width of $10w$ feet. Which expression represents the width, in inches, of the room in the model?

- A. $\frac{5}{8}w$
- B. $\frac{25}{8}w$
- C. $8w$
- D. $32w$

Answer Explanation

The scale is 5 inches in the model : 16 feet in reality.

Let the actual width be $10w$ feet. To find the width in the model, apply the ratio:

$$\frac{5}{16} = \frac{\text{model width}}{10w} \Rightarrow \text{model width} = \frac{5}{16} \cdot 10w = \frac{50}{16}w = \frac{25}{8}w$$

Correct Answer: $\frac{25}{8}w$

Thank You!

We appreciate your time and focus.

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